SUBA SIVA CHANDRAN KALIMUTHU

Python Backend engineer | AWS Cloud Engineer

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linkedin.com/in/subasivachandran | Personal Portfolio

PROFESSIONAL SUMMARY

Python developer with 6+ years in commercial engineering software development and earlier SAP ABAP experience. Delivered customer applications, SQL/ORM performance improvements, workflow automation and automated tests. Designed and deployed a multi-tier AWS application architecture with private EC2 compute, Application Load Balancer, RDS PostgreSQL, S3, Secrets Manager and Auto Scaling; built REST API endpoints using FastAPI.

TECHNICAL SKILLS

Backend & languages: Python 3, FastAPI, REST API development, SQL, PostgreSQL, ORM, OOP, SAP ABAP

AWS & infrastructure: VPC, EC2, ALB, Auto Scaling, RDS, S3, IAM, Secrets Manager, CloudWatch, Systems Manager, CloudFront, networking, security groups

Development & tooling: Git, GitHub, GitLab, Linux, shell scripting, Uvicorn, unit testing, integration testing, debugging, Matplotlib

PROFESSIONAL EXPERIENCE

GNS mbH | Python Developer / Simulation Engineer | Braunschweig, Germany | Mar 2022–Present

- Improved customer application data-management efficiency by 20% by developing a Python application using the gnsgui framework for CDH Japan.
- Reduced database query response times by 35% by optimizing SQL queries and ORM-based data access.
- Accelerated BMW-related automation scripts by 40% by migrating legacy TCL workflows to Python.
- Achieved 95% test coverage and reduced reported production errors by 80% through unit and integration testing.
- Improved analysis efficiency by 15% by developing a Matplotlib visualization tool for simulation data.

Technical University of Braunschweig — Institute of Joining and Welding Technology | Research Assistant | Dec 2018–Sep 2021

- Reduced simulation dataset analysis time by 30% by developing Python post-processing tools for ABAQUS.
- Cut simulation result extraction time by 50% by automating repeatable data-processing workflows.

Technical University of Braunschweig — Institute of Dynamics and Vibrations | Research Assistant | Jan–Mar 2019

- Improved impact-assessment accuracy by 15% through MATLAB script optimization; increased code reuse by 25% with reusable functions.

Tata Consultancy Services | Assistant Systems Engineer (SAP ABAP) | Bengaluru, India | May 2015–Sep 2017

- Improved ABAP program performance by 20% through code optimization and debugging.
- Resolved integration issues across IDoc, RFC and BAPI interfaces; developed SAP forms and SQL CRUD programs.

BACKEND & CLOUD PROJECTS

AWS Multi-Tier Application Infrastructure | AWS, FastAPI, PostgreSQL | Sep 2026

- Architected and deployed a VPC with public/private subnets, Internet Gateway, NAT Gateway, route tables and security groups; hosted FastAPI endpoints on private EC2 instances behind an Application Load Balancer.
- Configured RDS PostgreSQL, S3 and Secrets Manager; verified S3 writes, IAM-authorized secret retrieval and application endpoint connectivity.
- Created an AMI and launch template, configured EC2 Auto Scaling and CloudWatch scaling policy, and verified healthy ALB targets.
- Used AWS Systems Manager for private-instance access; audited and decommissioned resources after testing to control cloud costs.

Crash Simulation Results REST API | Python, FastAPI, Uvicorn

- Developed FastAPI routes for health/version checks and simulation-run creation and updates (GET /health, GET /version, POST /simulationruns, PUT /simulationruns/{id}).
- Defined typed request/response models for project name, solver, vehicle model and load case; executed and verified the API locally with Uvicorn.

ADDITIONAL PROJECT ARCHITECTURES

Serverless Event-Driven Processing | AWS Lambda, API Gateway, S3, DynamoDB |

- Designed a serverless workflow for HTTP ingestion, object storage and event-driven processing, with IAM-scoped service access and CloudWatch observability.
- Specified API Gateway and Lambda request flow, DynamoDB persistence, failure handling and deployment verification criteria.

Containerized Backend Platform | Docker, Kubernetes, AWS |

- Designed a container-based deployment architecture for a Python API, including Docker image packaging, Kubernetes Deployments and Services.
- Specified configuration management, health probes, rolling updates and scaling considerations for a cloud-hosted API.

Simulation Results Management API | FastAPI, PostgreSQL, SQLAlchemy |

- Designed CRUD resources for simulation runs and result metadata, with relational PostgreSQL schemas and ORM-based data access.
- Specified input validation, pagination, error responses and automated API tests for maintainable backend services.

Secure Job Processing API | FastAPI, Redis, PostgreSQL |

- Designed asynchronous job-submission and status endpoints, including persistent job metadata and queue-backed worker processing.
- Specified authentication, idempotent requests, retries and structured logging for reliable API operation.

EDUCATION & CERTIFICATIONS

M.Sc. Computational Sciences in Engineering — Technical University of Braunschweig, Germany (2021)

B.E. Mechanical Engineering — Government College of Engineering, Erode, India (2015)

AWS Certified Cloud Practitioner (CLF-C02) , Sep 2026 | Harvard CS50 Web Programming with Python and JavaScript

LANGUAGES

English (fluent) | German (fluent) | Tamil (native)